Constraining Dark Energy: Reducing Shape Noise of Shear Measurements using Kinematic Lensing

Weak gravitational lensing (WL) is a leading method of mapping the large-scale structure of the universe and its matter distribution. Precise measurements of galaxy shape distortions caused by WL, known as cosmic shear, enable tight constraints on the growth of cosmic structure and thus inform the physics driving the cosmic acceleration of the universe. However, the WL distortion of observed galaxies is 1% of the intrinsic ellipticities of galaxies, making it **impossible to differentiate** the WL signal from an unlensed galaxy. This issue is known as "shape noise". We propose to validate a promising alternative method of shear measurement called kinematic lensing (KL) using JWST grism data of spiral galaxies in the SAPPHIRES dataset. KL combines photometric and spectroscopic measurements to break degeneracies of intrinsic galaxy shapes, robustly differentiating sheared and unsheared galaxies, and **reducing shape noise by an order of magnitude**. By applying KL to the JWST SAPPHIRES dataset, we will validate its efficiency and effectiveness with slitless spectroscopy and perform analysis on the roll-angle dependency of KL. The upcoming Nancy Grace Roman Telescope (Roman), which has a similar grism to JWST, launches later this year. It is thus crucial to thoroughly test KL using data taken with JWST's grism to **refine and improve the method** for future Roman grism measurements. An empirically tested model of KL is critically needed to realize the promise of galaxy shape measurements to yield precision constraints on cosmic evolution and dark energy.

- ## Scientific Justification

Kinematic Lensing Will Solve the Issue of Shape Noise in Weak Lensing

Weak gravitational lensing (WL) is the **leading method to map the large-scale structure** (LSS) of the universe and its matter distribution. The minute distortions by WL, known as cosmic shear, provide direct insight into the LSS. By averaging over shapes of galaxies, shear measurements can be utilized to create shear maps which once inverted become the projected map, thus creating a map of the dark matter distribution of the universe (Refregier 2003). The method of building matter distribution maps through WL removes the need for assumptions of physical models of the universe. This is an extreme advantage over other methods of mapping LLS, which commonly require physical assumptions (Hoekstra et al. 2002). Precise measurements of cosmic shear enable tight constraints on the growth of cosmic structure and thus motivate the physics driving the cosmic acceleration of the universe.

However, WL introduces distortions that are up to 1% of intrinsic ellipticities, making it **impossible to differentiate** between lensed and unlensed galaxies. This issue is known as "shape noise," and the noise is an order of magnitude larger than the WL signal (Huff et al. 2019). Solving the issue of shape noise in WL measurements is **crucial for the effectiveness** of upcoming Stage IV surveys, like Euclid and Roman, that will provide massive amounts of data for lensing measurements and open the door for unprecedented precision cosmology results.

<u>We propose to validate our method of kinematic lensing (KL) using JWST SAPPHIRES spiral galaxies</u>. KL combines spectroscopic and photometric data to **alleviate the degeneracies** of cosmic shear and reduce shape noise by an order of magnitude using the spiral galaxies and multiple roll angles provided by the SAPPHIRES data set. Precise galaxy shape measurements from KL on JWST spiral galaxies will provide fresh opportunities to study galaxy shapes and the environments which created them, advancing the field further into understanding the process of galaxy evolution. With a new method of analyzing galaxy shapes, we will precisely and accurately determine the LLS map of the universe and tightly constrain cosmological models in the **era of precision cosmology**. KL further allows for new theoretical testing of physical theories and models of the universe using empirical galaxy shape measurements and shear maps.

Shape Noise dominates Weak Lensing measurements

The WL distortions of observed galaxies are on the order of 1% of the intrinsic ellipticities of galaxies, an issue known as "shape noise" (Niemi et al. 2015). However, this slight distortion makes it **impossible to differentiate** between lensed galaxies and intrinsically elliptical galaxies. The small distortions of galaxies require averaging over large samples of galaxies, on order of 10^2 or 10^3 galaxies per arcminute (Mendelbaum 2018). Reducing shape noise as a massive source of statistical error will allow for more precise individual galaxy shape measurements, without requiring substantial survey sizes. Accurate and precise measurements of galaxy shapes will enable tight cosmological constraints on physical models and modeling of LSS.

The method of KL will <u>robustly differentiate individual elliptical galaxies from lensed galaxies,</u> reducing shape noise in galaxy shape measurements by an **order of magnitude**. We will robustly verify KL's efficiency and accuracy using the JWST

SAPPHIRES spiral galaxies and firmly establish KL as an improvement to the current leading method of WL for mapping the LLS of the universe.

This Proposal: JWST SAPPHIRES Spiral Galaxies will Validate the Kinematic Lensing Model

The SAPPHIRES data release provides image cutouts and 2D spectra of spiral galaxies, taken by NIRCam and its grism (Sun et al. 2025). JWST provides galaxy images across a wide range of wavelengths, 0.6 to 29 µmeters, and slitless spectroscopy using the grism module on NIRCam, allowing for a uniquely powerful and insightful data set for KL (Gardner et al. 2006). JWST's slitless grism is similar to Roman's grism, making this data essential for the preparation of KL for galaxy shape measurements using Roman data (Greene et al. 2017; Gao et al. 2023). This proposal appeals to the JWST science goal "Assembly of Galaxies" by providing new, unique galaxy shape measurements (Gardner et al. 2006). The newfound confidence in accurate galaxy shapes will enable the study of intrinsic galaxy shapes, galaxy formation and the underlying physics governing galaxy mechanics.

We propose to combine JWST SAPPHIRES images and 2D spectra of spiral galaxies to thoroughly validate the use of KL to obtain precision galaxy shape measurements. Before applying KL, the data will undergo preprocessing to collect intensity information from the image cutouts and to remove the continuum from the 2D spectra. With the developing model, KL will be applied to the real, processed data using selected priors for galaxy features, including those featured in Figure 1, and the shear shape measurements. Within the model, as shown in the center of Figure 1, the combination of photometric and spectroscopic measurements will result in the major axis velocity of the galaxy. Appling the Tully Fisher Relation to the galaxy images to find the circular velocity and combining with the major axis velocity will provide the galaxy disc inclination (R.S. et al. 2023). Once the disc inclination is independently combined with the photometric measurements, KL will produce shear measurements for the spiral galaxies included in the JWST SAPPHIRES data with reduced shear measurements from WL by an order of magnitude. The resulting measurements will then be rigorously tested against stringent requirements and analyzed to validate the efficiency and accuracy of the KL method.

Using the JWST SAPPIRES spiral galaxies, the KL model will be rigorously tested, validating its use for further galaxy shape measurements. The KL model will allow for measurements of galaxy shapes with an order of magnitude less shape noise contribution to the statistical error compared to WL measurements. These measurements will enable unprecedented precision mapping of LSS and cosmological constraints.

Kinematic Lensing Will Pave the Way for the Era of Precision Cosmology

Within the era of precision cosmology, a validated and trusted KL model will provide unprecedented precise and accurate galaxy shape measurements with a reduced shape noise contribution to statistical error. Improvement upon pre-processing methods of spectroscopic and photometric data is only essential for KL but will also greatly benefit all science cases which require pre-processing. The succession of WL by KL as a **leading and reliable method** of mapping LSS will greatly advance our ability to study cosmic evolution and the underlying physics that drives it.

Upcoming Stage IV surveys, like Roman, will provide **massive, unprecedented amounts** of imaging and slitless grism data. Taking advantage of this data with the ability to **accurately, precisely, and efficiently analyze** this data using KL is essential. In alignment with JWST's science goal of "Assembly of Galaxies," robustly measuring the shapes of galaxies with KL allows for distinct opportunities to study the formation of numerous galaxies, determine the cause of their intrinsic shapes, and observe how nearby galaxies affect each other (Gardner et al. 2006). By observing numerous galaxies, especially those at high redshifts (z >10) according to JWST science goals, KL can be tested to the extremes of observation and used as a reliable method. The greater adoption of KL as a means of measuring galaxy shapes, as validated by SAPPHIRES spiral galaxies, could greatly advance the field of cosmology. Validation of KL will allow for application to realistic simulations and motivate future survey plans for JWST, Roman, Euclid and other upcoming missions. By improving statistical error by an order of magnitude and providing accurate galaxy shapes, KL will enable the field to test more physical models and establish tight constraints on physical parameters. Robust measurements within the SAPPHIRES sample will elucidate KL's ability to **perform powerful analysis and pave the way** for galaxy formation and cosmic evolution research in the era of precision cosmology.

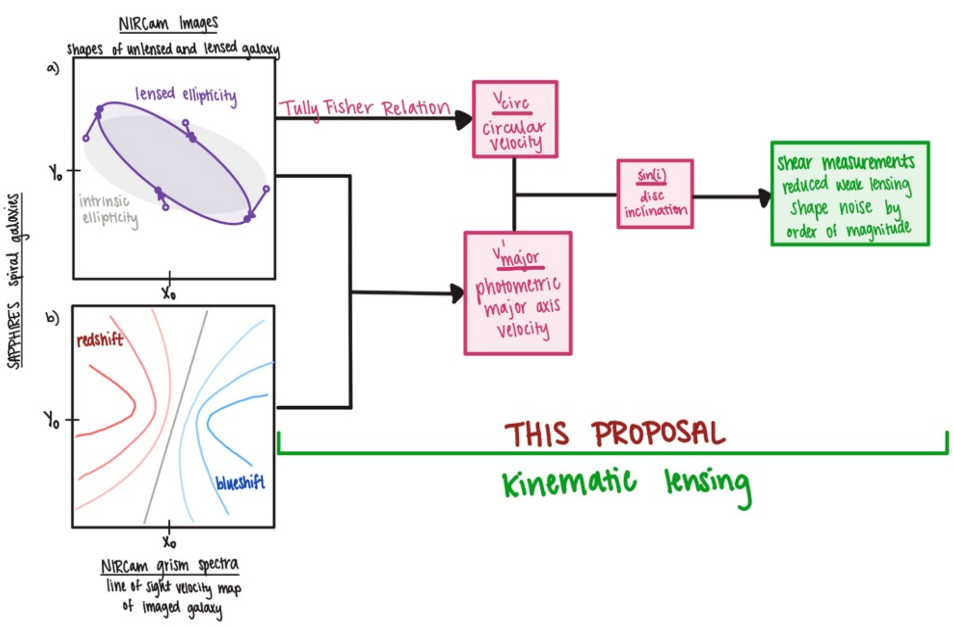

Figure 1. By combining photometric and spectroscopic JWST data, KL can significantly reduce shape noise and allow for precise shear measurements, paving the way to map the matter distribution of the universe in the era of precision cosmology. a) cosmic shear in JWST NIRCam photometric samples, illustrating an intrinsic galaxy shape (grey with open circles) changes due to lensing (purple with closed circles). b) NIRCam grism spectroscopic measurements as a line-of-sight velocity map, showing the redshift and blueshift of the galaxies. The lines and arrows depict the combination of data and products to produce shear measurements. WL alone (a) is severely limited by shape noise, a significant contribution to statistical error Huff et al. 2019). <u>We propose to apply the novel method of KL to the JWST SAPPHIRES spiral galaxies to demonstrate shape noise can be reduced by an order of magnitude, validating this method as a pathway to yield tight constraints on cosmic evolution and dark energy models</u>. This program will further calibrate the KL technique, setting the stage for future shear measurements using Roman's grism.

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